Rubin Rao

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Education

Bachelor of Science in Mechanical Engineering 3.50 | Aug. 2024 - May 2028
Texas A&M University *College Station, TX*

Objective Statement

Aspiring Mechanical Engineer committed to owning problems end-to-end across the full engineering lifecycle, from conceptualization to analysis and design to manufacturing. Seeking opportunities for the Summer of 2027.

Experience

Textron Systems - Uncrewed Land and Air Systems (ULA) Jun. 2026 – Aug. 2026
Mechanical Design Engineering | Internship *Hunt Valley, MD*

- Supported Aerosonde motor characterization test stand development through mechanical design and hardware integration
- Created CAD model and manufacturing drawings for a 15 kW generator test stand using NX
- Developed and released manufacturing drawings for various bracketry and assemblies using GD&T principles

Bell Flight Jun. 2025 – Aug. 2025
Manufacturing/Process Engineering | Internship *Grand Prairie, TX*

- Designed and iterated a rivet installation tool to resolve recurring quality defects on the 407 Tail Rotor Shaft Assembly
- Reorganized 150+ masking rubber types across two storage units, implementing a labeled catalog to improve traceability
- Collaborated with operators to document production grind setup and procedures across 15+ standard work operations

Texas A&M Sounding Rocketry Team (SRT) Aug. 2024 – Aug. 2026
Structures Engineer | Student Design Team *College Station, TX*

- Owned the design, analysis, and manufacturing of a modular gantry system capable of withstanding high dynamic loads
- Manufactured structural components using oven-cured preimpregnated composites, layups, and TIG Welding
- Presented and defended structural designs in 3+ formal Design Reviews

Projects

Aerosonde Motor Characterization Bracketry Jun. 2026 – Aug. 2026
Textron Systems - ULA *Hunt Valley, MD*

- Designed bracketry to mount sensors and motors to test stand for characterization of thrust, torque, and propeller speeds
- Developed milled aluminum adapter to interface between load cells and 2+ motor configurations
- Architected system to place Hall Effect sensor within 2 mm of encoder face, while withstanding vibrations produced by test stand equipment
- Leveraged Ansys to perform modal & structural analysis on bracketry to characterize & mitigate vibrational & loading effects

C.R.A.N.E. Aug. 2025 – Mar. 2026
Texas A&M Sounding Rocketry Team *College Station, TX*

- Developed a gantry system and tether system to transport a 400-lb Lander Vehicle, engineered to withstand 2,000 lbs equivalent dynamic loading
- Performed beam bending hand calculations to properly size Gantry System and Tether System hardware
- Conducted Finite Element Analysis (FEA) in Ansys Static Structural to identify potential stress concentrations in the mounting hardware under the 2,000-lb loading condition and drive solutions to redistribute mechanical stresses

Bell 407 Tail Rotor Blind Rivet Installation Tool Jun. 2025 – Aug. 2025
Bell Flight *Grand Prairie, TX*

- Identified operational root causes of blind-rivet installation defects on the Bell 407 tail rotor shaft
- Designed a tool system to constrain shaft movement and align the rivet gun normal to the rivet head, incorporating technician feedback to improve system ergonomics and accessibility
- Utilized Fusion360 and FDM 3D printing to rapidly iterate through four prototype generations, improving fit, function, and accessibility at each stage

Technical Skills, Involvement

- **Software:** Ansys (Static Structural, Modal) NX, Solidworks, Fusion 360, Microsoft Office Suite
- **Manufacturing:** Mill, lathe, composites (wet layup, pre-impregnated fiber), TIG welding, 3D Printing, laser cutting
- **Involvement:** Texas A&M Formula SAE - Internal Combustion, National Eagle Scout Association